Handout: from a loop to a datasheet

IEEE SENSORS 2026 tutorial, Sunday 25 October: From Naïve Loops to FAIR Datasheets

This side is for the talk. The other side is for home. We vote by hands up, or with the poll on your phone. The room chooses at the start. Poll: [poll link, if used].

1 The game: you are the logger

A logger writes down the values it sees. Write one number in each box. Write it only at the beep. Leave a box empty if you missed it.

Round 1

Number												

Round 2: write the frame number too

Frame number												
Number												

Round 3: two beeps for each new value

Beeps 1 to 12												
Beeps 13 to 24												

After each round, look at your strip and your neighbour's: find the numbers you skipped and the numbers you wrote twice.

2 Hands-on task: five minutes, in groups of two or three

Open the app: `uv run demo/app.py` in the repository folder. Install steps: see the back. No laptop? Join a neighbour.

1 Choose V1 (the version with one loop). Press Start with the short run. Read: 1 gap in 100 is longer than _____ ms

2 Choose V2 (the version with the queue). Do a short run again. Read the same number: _____ ms

3 Look at "Where the time goes". Which part moved out of the main loop? _____

Draw the five lanes and the stop column from memory.

3 The check box: eight questions for your own rig

During the talk, write each question in its box. At the end, write your answer on the right. A number needs its unit. If you do not know the answer, write a question mark: that is your homework. For yes or no, and for A to D, circle one.

	Question, in your words	Your answer
1		ms
2		yes no
3		ms
4		ms
5		ms
6		ms
7		A B C D
8		yes no

To take home

How to read 1 gap in 100, how to measure each check answer on your rig, and two details from the talk.

4 How to read “1 gap in 100”

1. Log the time of every row. The gap is the time of a row minus the time of the row before it.
2. Sort the gaps from the longest to the shortest. Read the value 1 in 100 from the top: with 1000 gaps, that is the 10th longest gap. You need at least 100 gaps.

5 How to measure each check answer on your rig

The answer is a number in ms, except questions 2 and 8 (yes or no) and question 7 (A to D).

1	**How long is 1 gap in 100 in your loop?**	Use the recipe above on a normal test of your rig.
2	**Can you count your lost frames?**	Write the frame number of the sensor in every row. Count the missing numbers and the repeated numbers. If the sensor value changes at every frame, count the repeated values instead. Yes, if you can count them.
3	**What else runs inside your loop?**	In one pass of the loop, read a clock before and after each call: the read, the file write, the plot, the rest. Write the time of the longest call.
4	**Which calls block, and for how long?**	A call blocks if the loop waits until it returns. Read a clock before and after each call that moves the stage or reads a device. Write the longest time. The manual of your stage says whether the call waits for the move.
5	**How long is 1 gap in 100 while plotting?**	Use the recipe twice: once with the plot open and once with it closed. Write 1 gap in 100 with the plot open.
6	**How long from STOP to standstill?**	Use a test with no specimen. Log the time of the STOP press. Log the time when the stage position stops changing, from the controller. Subtract. A software stop does not replace your hardwired emergency stop.
7	**Whose clock is in your file?**	Find the line that makes the time column. A: the sensor sends the time of the frame. B: the computer, when it read the frame. C: the computer, when it wrote the row. D: you cannot tell.
8	**Has your analysis run on a file with a known answer?**	Take a good file and plant a fault on purpose: for example, drop every third frame. Run your analysis on it. Yes, if it finds the fault.

6 Detail: Windows and a minimised window

A call such as `time.sleep(0.001)` asks for a wake-up after 1 ms. Windows decides when the program wakes up, and it follows its system timer. Since Windows 10, version 2004, each program asks for its own timer resolution (`timeBeginPeriod`). Starting with Windows 11, a program whose window is minimised, or fully hidden, gets no promise of a finer resolution than the default. So minimising the window of your rig can change the gaps in your loop. To check, read 1 gap in 100 once with the window open and once with it minimised.

Sources: docs.python.org, library/time; learn.microsoft.com, timeBeginPeriod, Remarks.

7 Detail: a database file that a viewer reads while you log

A database file (SQLite) with its write-ahead log (WAL) switched on lets a viewer read the rows while the test runs. The switch is `PRAGMA journal_mode=WAL;`. Only one program can write to the file at a time. A viewer sees the rows that the writer has saved: if the writer saves once a second, the viewer is up to 1 s behind. Keep the file on the same computer as the program that writes it. Network drives do not work.

Source: sqlite.org, wal.html.

The demo, this handout and the sources: `github.com/matteolopreti/sensors2026-tutorial`.

The page has install steps for Mac, Windows and Linux.